First session of International Theoretical & Practical Hands-on Course on

"Using whole-genome data in plant breeding: GWAS and Genomic Selection to genetically characterize key traits in large germplasm collections"

23-27 July, 2018, VIR, St Petersburg

Day 1. How to make the best of large-scale phenotyping raw datasets : Focus on some useful experimental designs and biostatistics with R

Case study #1: Randomized Complete Factorial Design applied to plant breeding.

Objectives: Test for effect of variety and mode of plant cultivation on drought tolerance using fixed linear model. This simple example also aims at familiarizing with R software.

Context: In the frame of a citrus breeding program, 3 varieties of lemon trees were cultivated in 3 shady conditions to test for drought tolerance. The 'Leave length/thickness ratio' character (named 'index' in the dataset '01_Exmpl1_Citrus.csv') was measured as a proxy functional trait for drought tolerance. We hypothesize that there is no interaction effect between lemon varieties and shady conditions for the 'Leave length/thickness ratio' character.

1. Describe the experimental design implemented for this study.

2. Are the tested varieties different in terms of leave length/thickness ratio?

3. Is there an effect of the shade level on the leave length/thickness ratio?

Case study #2: RCBD applied to genetic analysis of quantitative traits (Genetic values, heterosis, ploidy).

Objectives: Key parameters for quantitative genetic analysis, such as genetic values of parental and F1 hybrid genotypes, heterosis, and effect of ploidy will be presented, calculated and interpreted as part of a breeding program.

Context: In the frame of a watermelon breeding program, two varieties were crossed to improve sugar content. In order to evaluate the interest of crossing these two varieties, we compare the performances of the parental varieties and F1 progenies. For this, the sugar content was evaluated during a greenhouse trial under semi-controlled environmental conditions (Datafile: 02_RCBD_SugarPloidyWatermelon.csv). We hypothesize that there is no genotype-environment interaction effect for the 'sugar content' character.

1. Describe the experimental design implemented for this study.

2. Are the parental varieties genetically different in terms of sugar content control? Conclude as to the advantage of crossing these two varieties to initiate a breeding program.

3. Is there an over-dominance effect in genetic control of sugar content ?

4. Among the three tested varieties, which one would you market? What are its advantages and drawbacks in terms of production, cultivation and commercialization?

Case study #3 (by yourself): RCBD applied to phytopathology.

Objectives: Implement by your own the concepts on experimental design and statistical models and the methods for biostatistics as developed in previous case studies.

Context: The fungus *Microdochium nivale* (Fr.) Samuels & Hallett is the most widespread snow mould fungus. It is distributed over a wide range of temperatures, but is regarded as a serious pathogen mainly in cold to temperate regions of the Northern hemisphere. *M. nivale* causes pink snow mould on turf and forage grasses, and winter cereals. At warmer temperatures, it causes stem rot, leaf blotch, and "Fusarium ear blight".

Wheat is among the main hosts of this pathogen. Ten winter wheat cultivars were assessed for components of partial disease resistance. In the glasshouse experiment, at GS 65 (mid-anthesis) (Zadoks et al., 1974), the reference isolate of *M. nivale* was inoculated against each cultivar (two heads per plant). Treatments were replicated four times for each cultivar. Visual assessment of disease levels was carried out on treated heads at GS 70, 80 and 90 (the percentage of infected spikelets per head). Relative Area Under the Disease Progression Curve (AUDPC) was calculated (Shanner and Finney, 1977) (Datafile:03_RCBD_AUDPCWheatFEB.csv).

1. Describe the experimental design implemented for this study.

2. Using a graphical representation of the data followed by a statistical analysis, conclude on the effect of the cultivar on the M. nivale head blight disease severity.

3.What does this mean for the breeding program and for varieties marketing?

Case study #4: Augmented-block designs for large-scale genotype collection evaluation: Evaluation trials for new varieties registration.

Objectives: Implementation of augmented block designs as well as advantages and limits of such experimental designs will be discussed. Bio-statistical analysis using R software will be presented.

Context: As part of new seed variety evaluation and French registration process, official trials are conducted and approved by the French GEVES Institute (Union for Plant Variety Studying and Seeds). Varieties are evaluated regarding their agronomic, technological and environmental value, also known as V.C.U.S (Value for Cultivation Use and Sustainability) trials. Protocols are those approved by the French CTPS Institute for variety assessment of cereals, rapeseed, sunflower... The studies focus on yield, yield stability factors (disease resistance, climate resistance, yield components) and variety use performance after at least two years of observation.

These trials are conducted on plots specifically selected regarding their agro-climatic characteristics and overall value for agronomic experiments.

In this study, the weights of eight new varieties of oats are compared to performances of four reference controls (Checks) in an augmented-block design with three blocks (Datafile: 04_AugmentedBlockDesign_example1.csv).

1. Which are the varieties that significantly outperform the reference controls?

Case Study #5 (by yourself): *Screening of a core collection of 30 accessions of common bean for high yield.*

Context: Because of the continuous introduction of germplasm from abroad, some collections have a high number of accessions, making it difficult to explore the genetic variability present in a germplasm bank for conservation and breeding purposes. Therefore, Perseguini et al. (2015) quantified and analyzed the structure of genetic variability among 500 common bean accessions to construct a core collection. A total of 58 SSRs were used for this purpose. The 500 accessions were clustered into 15 distinct groups and the 180 accessions into four distinct groups in the Structure analysis. According to analysis of molecular variance, the most divergent accessions comprised 97.2% of the observed genetic variability present within the base collection, confirming the efficiency of the selection criterion. The 180 selected accessions will be used for association mapping in future studies and could be potentially used by breeders to direct new crosses and generate elite cultivars that meet current and future global market needs.

The yield of thirty accessions equally distributed within the four genetic groups was compared to performances of four reference controls (Checks) in an augmented-block design with six blocks (Datafile: 05_AugmentedBlockDesign_exemple2.csv).

1. Which are the genotypes of utmost interest to initiate a breeding program to improve yield in common bean ?

Perseguini JM, Silva GM, Rosa JR, Gazaffi R, Marçal JF, Carbonell SA, Chiorato AF, Zucchi MI, Garcia AA, Benchimol-Reis LL. Developing a common bean core collection suitable for association mapping studies. Genet Mol Biol. 2015 Mar;38(1):67-78. doi: 10.1590/S1415-475738120140126. Epub 2014 Mar 17

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